

ELECTRONIC ANSWER BOOK

I AGREE TO ABIDE BY THE SENATE POLICY ON CONDUCT OF ASSIGNMENTS, MID-TERM EXAM, AND FINAL EXAM

Instruction: Using Microsoft Word/WordPad/Notepad, type Student's name, Student ID number, type Chapter title and applicable Question Number for Assignment, Mid-Term Exam or Final Exam (as listed under Brightspace, Quizzes) before typing Solution below. One separate file for each Assignment/Mid-Term/Final Exam Question.

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Midterm – AM – Question: 2a

a) Weight Matrix W –

$$X_1^T Y_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} [1 \quad 1 \quad -1 \quad -1]$$

$$= \begin{bmatrix} 1 & 1 & -1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & 1 & -1 & -1 \end{bmatrix}$$

$$X_2^T Y_2 = \begin{bmatrix} 1 \\ -1 \\ 1 \\ -1 \end{bmatrix} [1 \quad -1 \quad -1 \quad 1]$$

$$= \begin{bmatrix} 1 & -1 & -1 & 1 \\ -1 & 1 & 1 & -1 \\ 1 & -1 & -1 & 1 \\ -1 & 1 & 1 & -1 \end{bmatrix}$$

$$W = X_1^T Y_1 + X_2^T Y_2$$

$$= \begin{bmatrix} 1 & 1 & -1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & 1 & -1 & -1 \end{bmatrix} + \begin{bmatrix} 1 & -1 & -1 & 1 \\ -1 & 1 & 1 & -1 \\ 1 & -1 & -1 & 1 \\ -1 & 1 & 1 & -1 \end{bmatrix}$$

$$= \begin{bmatrix} 2 & 0 & -2 & 0 \\ 0 & 2 & 0 & -2 \\ 2 & 0 & -2 & 0 \\ 0 & 2 & 0 & -2 \end{bmatrix}$$

b) Noisy vector:

$$Y_t = [1 \quad -1 \quad 1 \quad 1]$$

$$G_h[Y_t W^T] = G_h[[1 \quad -1 \quad 1 \quad 1] W^T]$$

$$= G_h[[0 \quad -4 \quad 0 \quad -4]]$$

$$= [1 \quad -1 \quad -1 \quad -1]$$

$$= X_2$$

$$G[X_2 W] = Y_2$$

The steady-state recall pattern pair is X_2 and Y_2 .

Sum of all the elements of $W =$

$$= \begin{bmatrix} 2 & 0 & -2 & 0 \\ 0 & 2 & 0 & -2 \\ 2 & 0 & -2 & 0 \\ 0 & 2 & 0 & -2 \end{bmatrix}$$

$$= 2 + 0 + -2 + 0 + 0 + 2 + 0 -2 + 2 + 0 -2 + 0 + 0 + 2 + 0 -2$$

$$= 0$$